Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Reduced phase rule, phase diagram of two-component system - Pb-Ag system

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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M1

Reduced phase rule, phase diagram of two-component system - Pb-Ag system

1 Reduced phase rule, phase diagram of two-component system - Pb-Ag system

Study focus: Reduced phase rule, phase diagram of two-component system - Pb-Ag system. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000202 · EV0000203

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module three on phase equilibria and in today's class we are going to learn how to construct a phase diagram of a two component system in general and specifically the lead silver system.

Key details

- Then we study the determination of solid liquid equilibria and finally the Patterson's process.
- When we talk of a two component system, we know that we need to use the phase rule F is equal to C minus P plus 1, which is known as the reduced phase rule.
- It is known as the reduced phase rule because we have reduced the degree of freedom by one.
- Is equal to one, then the degree of freedom becomes equal to three, and we need to draw a three dimensional phram.
- Most often we use pressure as constant variable.
- So therefore because we reduce the degree of freedom by one, the reduced phase rule is now applicable.
- F is equal to c minus p plus one.
- And now out of pressure, temperature and composition, since pressure is being maintained constant, and especially for solid liquid equilibria, pressure is chosen constant variable.
- When we are drawing this phase diagram of a two component system, this particular phase diagram will be known as a simple eutectic system.
- One temperature scale stands for pure A and the other temperature scale stands for pure B.
- And on the x axis we have the composition scale.

- What is happening is here we have 100% A and as we move along this axis we are adding B into A.
- of various compositions which are represented by A and B, the percentage of A and B.
- Now the first thing we need to know is the melting point of pure A.
- Now that melting point because this axis represents pure A and the temperature scale.
- the alphabet M.
- Now as we go on adding B into A, the we are actually adding an impurity into the pure substance A, and therefore we know that there is going to be a depression in freezing point.
- When we add a certain amount of B, say a small amount of B, we find that it reduces the freezing point of pure A.
- And as we go on adding more and more of A, we find that this goes on decreasing like this.
- So therefore if I have to join these points I get a plot like this.
- And now if I add A into the pure B again I will notice that there is a decrease in the melting point and if I join these again I get a line like this.
- If I join these two they happen to meet at a point which is known as the utectic point.

Example or application from the lecture

- And for that, the criteria is that suppose we have two components A and B, then A and B must be miscible in all proportions in the molten state or the liquid state and in the solid state they are completely immissible.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

Core explanation · 05:00–09:59

If we draw a phram, then this is what the phram is going to look like.

Key details

- For that, what we do is the simplest way is to take any composition of A and B, mixture of A and V, and we heat it.
- beyond this point.
- So that means if I go beyond this point, then the liquid is what we are going to get is the liquid melt.
- So therefore we have the liquid melt now.
- What I do now is I start cooling it.
- When I come to this point here,.
- I find that we have that I have reached the melting point or the freezing point of A in the mixture.
- The first crystal of A will start forming here.
- Remember that this is not pure A, it is a mixture, and in a mixture now the melting point has reduced or the freezing point has reduced.
- As we go further down.
- We find that more and more of this.
- A is going to freeze out of the mixture.

- That's exactly what happens here, and how do we know what is the.
- Liquid melt now it is at this point say if.
- See, it has become richer in B.
- So as we go down further, we go on cooling it, more and more of A will precipitate out and therefore.
- Liquid melt has now reached this composition.
- When we reach this point of the graph, what we notice is that the liquid melt has now reached this point.
- We have reached even the freezing point of B.
- Therefore, we will find that B also starts precipitating at this point.
- The liquid is cooling initially and then we get the first crystal of A.
- As we cool down further, more and more A separates out and the liquid melt becomes richer and richer in B.

Example or application from the lecture

- More of A has come out, we find that the liquid melt can be given by this point.
- Let us now label each of these areas.

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EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 10:00–14:59

And both of them will meet at the same utactic point E.

Key details

- Because all these points represent the melting points of the mixtures.
- Therefore, this whole area will be the liquid melt.
- We find here that we have the liquidment as well as the solid A because in this area what is happening?
- Solid A is being frozen out and precipitating out and we have the liquid melt.
- So this area would represent liquid melt plus the solid A and similarly this area would represent the liquid melt plus solid B because if you are cooling on this side we will find that B starts respirating out.
- Liquid mill which again is has both A and B and here we have the solid B in this area plus the liquid mill.
- F E G.
- We have both solid A as well as solid B because below this point what do we observe?
- The liquid melt now completely converted into the solids A and B and here we see there is a.
- Apart from this, the another point of interest which I have been mentioning is the point E which is known as the eutectic point and also these curves are of interest to us the curve ME and the curve NE.
- found to be a eutectic mixture.
- It's a mixture of the two components A and B, which has the lowest freezing point of all the possible mixtures of the components.

- Here the name eutectic is from a Greek word which means easily melting.
- And since has the lowest melting point amongst all mixtures of A and B, this is referred to as the eutectic mixture.
- One is the liquid phase, one is solid A, and the other is solid B.
- Freedom at this point.
- Therefore F comes out to be zero.
- So therefore this point is invariant.
- Now we know that when we have an invariant point, it means that we cannot change the temperature or the composition.
- And one more important point which we need to understand here is that the melting point is very sharp.
- unlike other mixtures where the melting point is over a range of temperatures, in the case of a eutectic mixture it is found that it has a very sharp melting point.
- In fact, earlier it was assumed that it is actually a compound, because compounds usually have a characteristic melting point, but later on microscopic examinations and other evidence showed that this.
- Temperature, this composition is actually that of a mixture.
- It is not a compound, it is a mixture, hence the name eutectic mixture, but it has a very sharp melting point.

Example or application from the lecture

- Let us first talk about the eutectic point which is shown by the point E here on the graph.

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EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 15:00–19:59

The important features of the utected point is that the mixture has the lowest melting point amongst all other compositions, then the it is an invariant point and also it has a very sharp melting point.

Key details

- The melting point of pure lead is 961 degrees centigrade that would be represented here, and that of pure lead is 327 degrees centigrade.
- And the eutectic temperature and composition become fixed and the eutectic temperature is found to be.
- To be 303 degrees centigrade, and the eutectic composition is 97.4% of lead and 2.6% of silver.
- Now this ME curve first is known as the freezing point curve of A.
- I have already explained that as we go on adding more and more of B, the melting point or the freezing point of A is decreasing.
- A and it also shows that as we go on adding B into A, the freezing point or the melting point is decreasing.

- Here we have solid A which is in equilibrium with the liquid melt which has which has B in A.
- And now the number of components are 2, the degree the number of phases is also 2, therefore the reduced phase rule is applicable.
- F is equal to C minus P plus 1 is equal to so 2 minus 2 plus 1 and therefore now it is 1.
- F is equal to 1 along this curve and therefore it is a univariant system.
- A into B again here we have on the curve solid B in equilibrium with the liquid melt which is now A added to B.
- And again we will use the reduced phase rule we find that.
- F is equal to 1, so the system is univariant along this curve.
- Thermal analysis is the study of what we call as the cooling curves of various compositions of a system during solidification.
- This curve, whatever curve we get here when we study the temperature versus time.
- We can determine the freezing point of the component in the mixture and also the euvectic point.
- How to construct the phase diagram using cooling curves.
- We have studied the simple eutectic system, so here we can see that the phase diagram is live like this.
- We will take up different compositions of A and B and try to plot the cooling curve for each of these compositions.
- We the initially if we when we when it is in the liquid form and we start cooling it, there is just cooling will happen and we don't see anything specific, but when it reaches the melting point or the freezing point, then solid A starts to form.
- And we know that at the freezing point the temperature gets arrested till.

Example or application from the lecture

- Now a very common example for a simple eutectic system would be the lead silver system.
- Let us understand these curves ME as well as NE.

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EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 20:00–24:59

all the liquid has got converted into solid and that is what we see because this is the this is the time axis.

Key details

- And then again when all this liquid has got converted into solid, we find that again cooling is going to happen.
- Say this is the composition and what are we doing here for cooling curve?
- We take this composition heat it to a high temperature beyond the.
- A high temperature when we see that the liquid now the mixture is in the liquid form, it's in the melt state.
- Then we start cooling it.

- When we cool it initially, we don't see anything, there is just a reduction in temperature.
- But the moment it reaches this point.
- We find that the first crystal of A will start to form.
- And we know whenever there is crystallization, then there is a latent heat of fusion involved.
- Definitely there is going to be a change in the slope and that is where we find a change here and that can be taken as the freezing point of A in the mixture B.
- Then further it keeps on cooling down till we come to the eutectic point.
- So when we come when we are cooling this we reach the eutectic protective point we know that now the temperature again gets arrested because now B is also being precipitated.
- We have three phases in equilibrium, F becomes equal to zero as we learned just now, and therefore the temperature becomes constant.
- This is known as the eutectic temperature.
- Which contains A and B of the same composition as the eutectic mill.
- Then after that, there is just a reduction in the temperature, that is there is just cooling.
- This break will tell us what is the freezing point of A in when B is added to it and this point also tells us the eutectic temperature.
- Similarly, if we take another composition, say this composition here, we find that when we melt it and start cooling it, now the freezing will happen at a lower temperature because we are we are here we have added more of B.
- If you look here, this is we have added.
- Added more of B into this, and therefore the freezing point will be much lower, and that's exactly what we see here.
- The first break now when the.
- A pure A will now start coming out of the system, starts precipitating out, we will get the freezing point of A in the mixture, and now this will happen at a much lower temperature because the freezing point has decreased further due to more of B in the mixture.
- Then after that, because of this freezing of A, we find a change in the slope, and that is what helps us to determine the freezing point.
- And then further, when we go down again, we see an arrest in the temperature, and that is the eutectic temperature.
- Composition or discomposition, the eutectic temperature is always constant.
- Further, there is just cooling.
- What is happening here?
- The A now is changing is crystallizing, it is changing from the liquid to the solid, so there is a.
- Arrest in temperatures till all of the liquid has got converted to solid and then there is cooling.
- And then coming to the next point, we have added more of A into B, and therefore we find there is a decrease in the freezing point that.
- Can be noted from the cooling curve where the first crystal of B starts appearing, there is a change in the slope.
- The slope was like this, now suddenly there's a change in the slope.
- Wherever that inflection point, that change in the slope happens, that will be the freezing point of B.

Example or application from the lecture

- Now let us take up the next composition.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

Core explanation · 25:00–29:59

when A is added into the mixture and then further cooling comes when it reaches the eutectic point.

Key details

- Again, so whether you approach from this side or you approach from this side the eutectic temperature.
- You are going to get the same value of utectic temperature.
- And similarly, if we take this now the eutectic point itself, that would be very interesting because what would happen now when you take this eutectic mixture and you heat it, we find that.
- Initially nothing happens, there's just cooling and the temperature just keeps falling down till we reach this point.
- The because we are at the eutectic point A and B both will start precipitating out and therefore we have three phases and this behaves so if you behave like a pure compound and we get the temperature arrest over here till the eutectic melt has completely converted into the two solids and the composition remains the same.
- The eutectic melt and the solids which are formed have the same composition.
- Therefore we find this arrest in temperature and then there is for the cooling.
- But here we find that there is you can know the utectic temperature.
- And we already know what is the eutectic temperature because that remains constant throughout.
- we can find out wherever these two lines curves meet that would be the eutectic composition.
- The lead silver system shows a simple eutectic diagram and in metallurgy there is a process known as the Pattenson's process.
- Which is used for desilverization of Argentiferous lead.
- The process of heating Argentiferous lead containing a very small quantity of silver, about 0.1%, and cooling it to get pure lead and a liquid melt which is richer in silver.
- When we start cooling it, when we reach this point here, we find that pure lead will start precipitating out and as we cool it further more and more of lead will precipate out and the liquid is going to get richer in silver.
- We go further cool it and then we see that more and more of the lead will precipitate out and the liquid melt composition now is moving towards the area where there is more of silver.
- The liquid melt is getting richer in silver.
- The lead now that is separating out floats and it is continuously removed by layers.
- The liquid melt now contains 2.6% of silver.

- to a composition where the liqu of the liquid weld which has which is much richer in silver and it has now the composition is two point six percent of.

Example or application from the lecture

- Let us now talk about an application of the phase diagram of a two component system, the lead silver system.

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EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 30:00–34:59

Natural ore which contains such a small quantity of silver can be made richer in the percentage of silver.

Key details

- So, in the next class we are going to talk about a more complicated system, that is the iron carbon system.

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EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00043	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00045	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

C Revision and retrieval practice